SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets.

QUESTIONS: 05 **Total Marks:50**

Annexure A

MOCK INTERVIEW

Criteria	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	Communication (1.5)	Confidence (1)	Response to Questions (1)	Total (10)
Weightage	25%	25%	15%	15%	10%	10%	100%
Q1							
Q2							
Q3							
Q4							
Q5							

Code of Conduct:

I KONGARA JOGESH(192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Jogesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25%	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25%	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15%	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15%	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15%	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10%	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Source note: the uploaded assessment shows Confidence as 10% on the Assessment Tool 3 front page, giving a 100% total, while the detailed rubric page lists Confidence as 15%. The report preserves the source wording and uses the front-page 100% distribution for the assessment mapping: 25%, 25%, 15%, 15%, 10%, 10%.

Annexure A — MOCK INTERVIEW: SAMPLE QUESTIONS

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question: Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question: Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question: Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4. Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question: Explain multilevel association rules using the dataset and compare them with multidimensional association rules.

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question: Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

CSA16 CO5 — MOCK INTERVIEW ASSESSMENT

Q1 — Retail Transactions

Direct Interview Answer

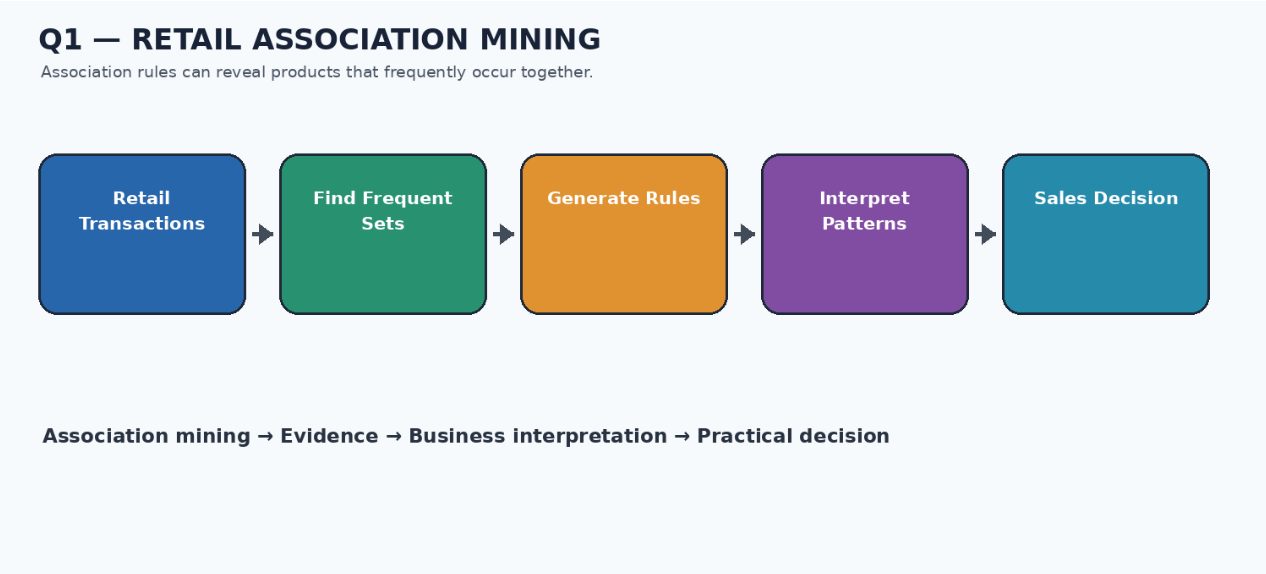
The dataset contains Milk, Bread, Butter and Eggs across five transactions. Association rule mining can identify products that repeatedly occur together. The retailer can use those relationships to improve product placement, recommendations and promotional bundles.

Dataset-Based Analysis

From the data, Milk and Bread occur together in T1 and T4, so their joint occurrence is 2 out of 5 transactions. Milk and Butter occur together in T2 and T4, while Bread and Butter occur together in T3 and T4. These recurring combinations indicate that customers sometimes purchase these products together. A practical retail decision would be to place complementary products closer together or recommend one item when another is selected.

Practical / Analytical Example

Example: if a customer selects Milk, the store can recommend Bread or Butter as a related purchase. The recommendation should be based on sufficiently frequent and reliable rules rather than on a single transaction.



Likely Follow-Up Response

Q: Why is this useful? A: It converts transaction history into actionable relationships that can support cross-selling, recommendation and store-layout decisions.

Rubric Evidence

Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	Response
Correct terminology and concept	Connects dataset to decision	Explains why the result/method matters	Structured direct answer	Professional wording	Includes relevant example

Q2 — Apriori vs FP-Growth

Direct Interview Answer

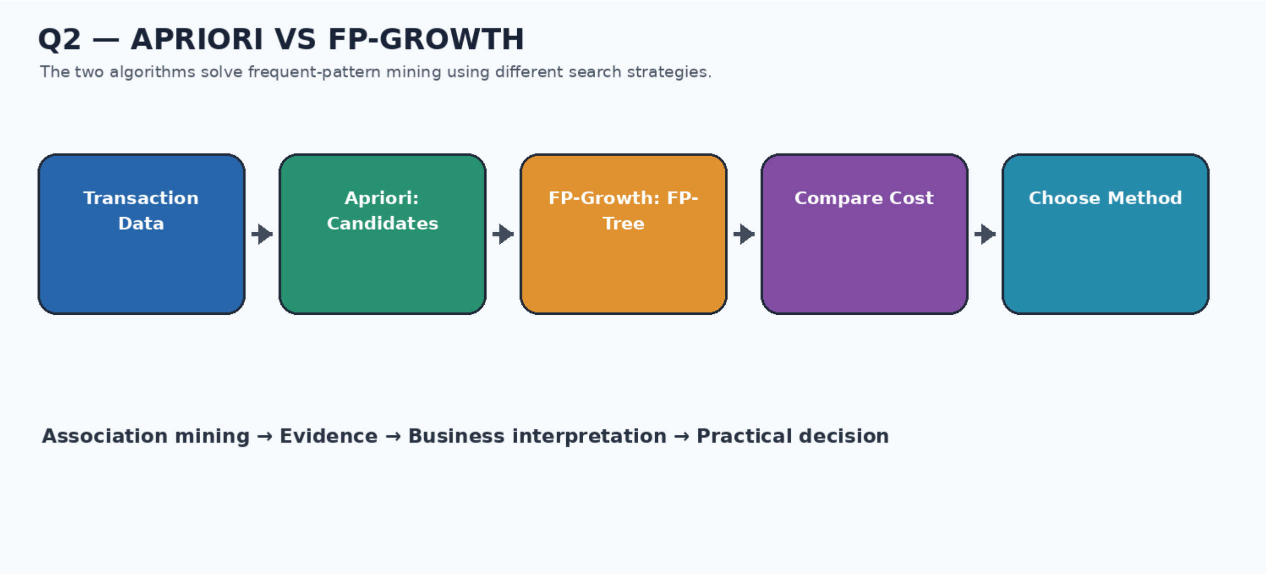
Apriori and FP-Growth are both frequent-pattern mining approaches, but they search for patterns differently. Apriori works level by level and generates candidate itemsets, then counts support and prunes candidates that do not satisfy the threshold. FP-Growth avoids explicit candidate generation by compressing transactions into an FP-tree and mining conditional pattern bases.

Dataset-Based Analysis

For the supplied transactions, Apriori is easy to explain and is useful when the dataset is relatively manageable and transparency of candidate generation is important. FP-Growth becomes attractive when candidate generation would create a large search space or repeated database scans become expensive.

Practical / Analytical Example

The key analytical difference is candidate handling. Apriori explicitly creates candidates such as {A,B}, {A,C}, {B,C}, and potentially larger candidates. FP-Growth stores shared transaction structure and mines that compressed representation.



Likely Follow-Up Response

Q: When would you choose FP-Growth? A: I would prefer it when the transaction database is large or contains many repeated patterns, because avoiding explicit candidate generation can reduce the search burden.

Rubric Evidence

Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	Response
Correct terminology and concept	Connects dataset to decision	Explains why the result/method matters	Structured direct answer	Professional wording	Includes relevant example

Q3 — Support and Confidence

Direct Interview Answer

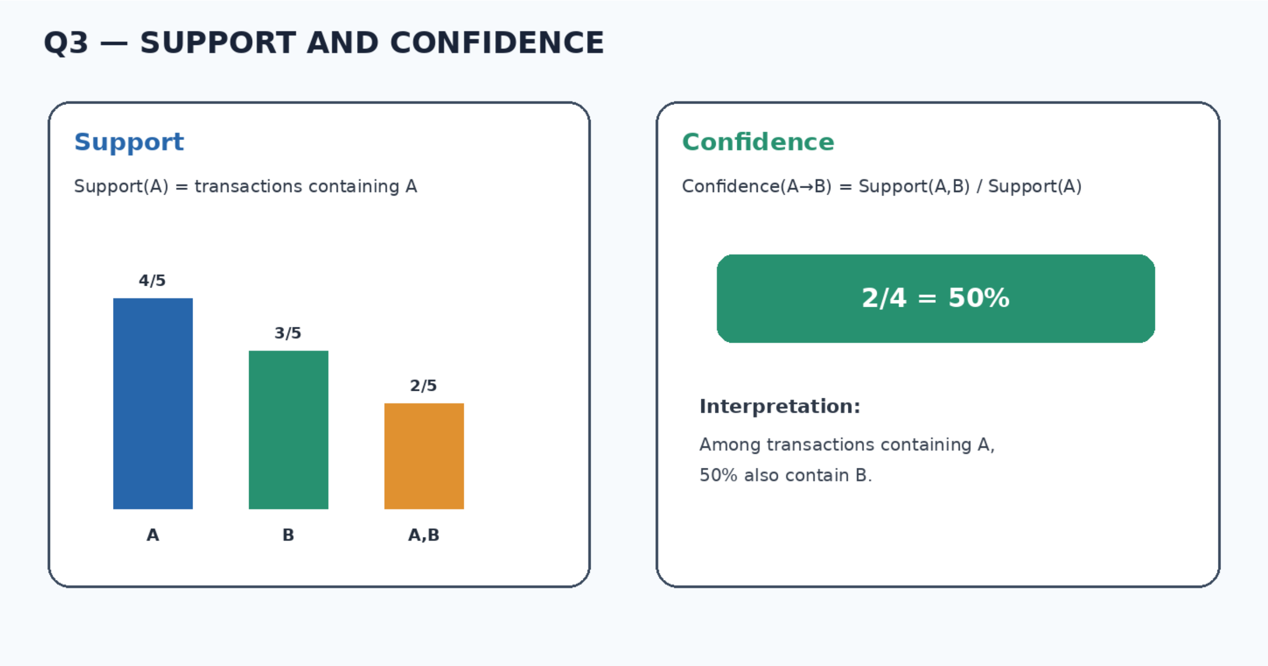
Support measures how frequently an itemset occurs in the complete transaction database. Confidence measures how often the consequent appears among transactions that contain the antecedent.

Dataset-Based Analysis

In the given dataset, A occurs in T1, T2, T4 and T5, so Support(A)=4/5=80%. B occurs in T1, T3 and T4, so Support(B)=3/5=60%. The combination A,B occurs in T1 and T4, so Support(A,B)=2/5=40%.

Practical / Analytical Example

For the rule A→B, Confidence(A→B)=Support(A,B)/Support(A)=2/4=50%. Therefore, 40% describes how common the combination is in all transactions, while 50% describes how often B appears when A is present.



Likely Follow-Up Response

Q: Why do both matter? A: Support helps remove very rare patterns, while confidence evaluates the reliability of a directional association. A rule should normally be considered using both measures together.

Rubric Evidence

Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	Response
Correct terminology and concept	Connects dataset to decision	Explains why the result/method matters	Structured direct answer	Professional wording	Includes relevant example

Q4 — Multilevel Association Rules

Direct Interview Answer

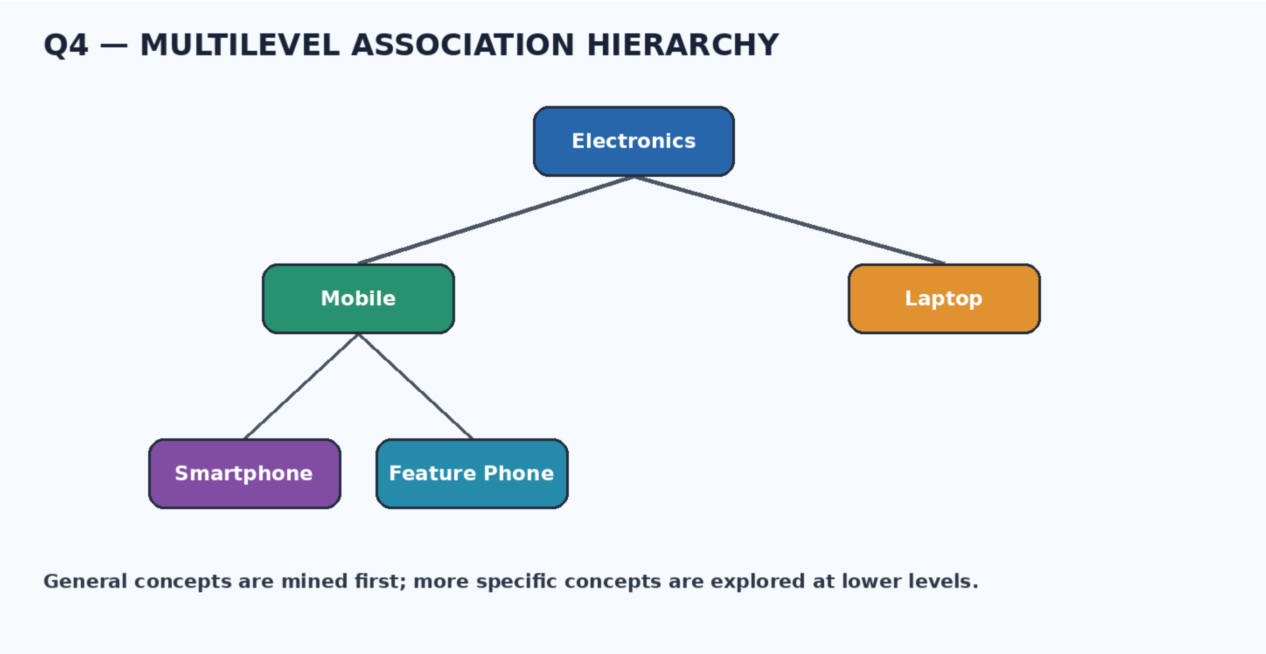
Multilevel association rules use a concept hierarchy so patterns can be discovered at different levels of abstraction. In the given data, Electronics is a general concept, while Mobile and Laptop are more specific concepts.

Dataset-Based Analysis

Electronics occurs in all five transactions, so its support is 100%. Mobile occurs in T1, T3 and T4, giving 60%. Laptop occurs in T2 and T5, giving 40%. This illustrates how a general concept can have stronger support than its more specific descendants.

Practical / Analytical Example

Compared with multidimensional association rules, multilevel rules focus on different abstraction levels within a concept hierarchy. Multidimensional association rules combine different attributes or dimensions, such as product category, customer age, location or purchase time. The two ideas can also be combined in a data-mining system.



Likely Follow-Up Response

Q: Why is the hierarchy useful? A: It lets an analyst first discover broad patterns and then drill down into specific concepts only where the data supports them.

Rubric Evidence

Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	Response
Correct terminology and concept	Connects dataset to decision	Explains why the result/method matters	Structured direct answer	Professional wording	Includes relevant example

Q5 — Mining Large Transaction Samples

Direct Interview Answer

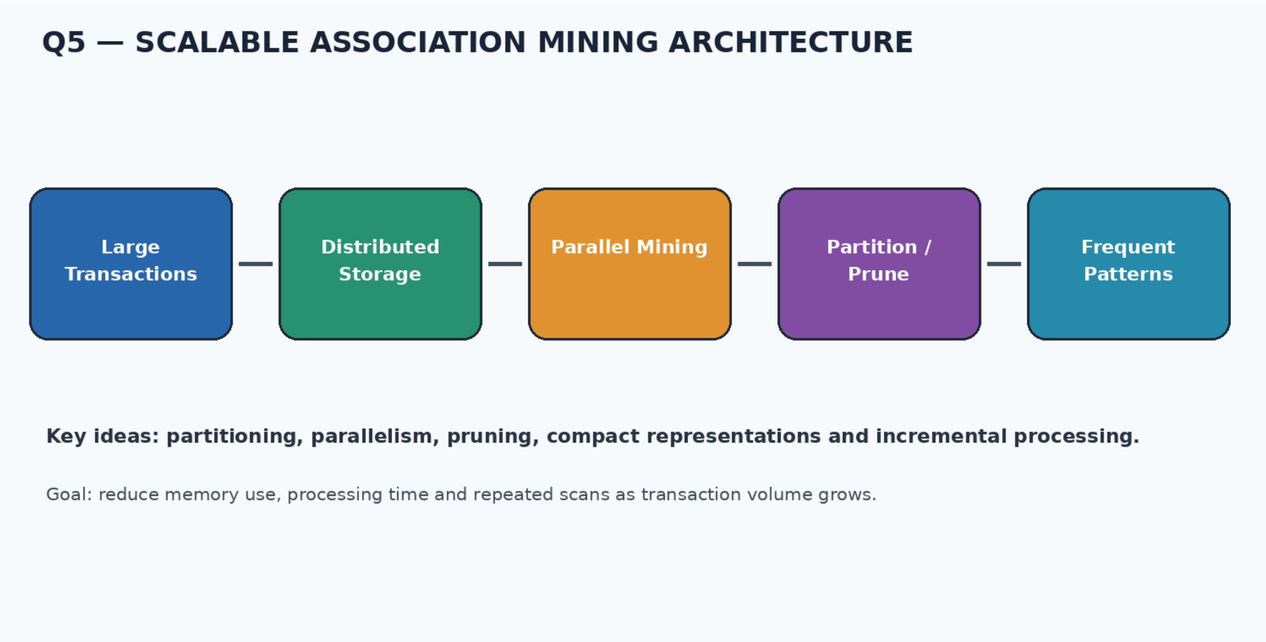
Mining large transaction datasets creates challenges in processing time, memory usage, storage, repeated database scans and the number of possible candidate patterns. Even a small sample such as the supplied five transactions demonstrates how combinations of A, B and C create multiple candidate itemsets.

Dataset-Based Analysis

A scalable solution can partition the dataset, process partitions in parallel, prune candidates early, use compact representations such as FP-trees, and reduce unnecessary database scans. Distributed processing can divide large transaction collections across multiple machines while preserving the overall mining objective.

Practical / Analytical Example

The main analytical principle is to reduce the amount of work before expensive pattern counting. If an itemset is clearly below a support threshold, it should not be carried into later stages. Parallelism and partitioning can then distribute the remaining computation.



Likely Follow-Up Response

Q: What is the benefit of scalability? A: It allows the mining method to remain practical as the number of transactions and candidate patterns grows, rather than increasing computation and memory requirements without control.

Rubric Evidence

Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	Response
Correct terminology and concept	Connects dataset to decision	Explains why the result/method matters	Structured direct answer	Professional wording	Includes relevant example

COMPARATIVE SUMMARY

Question	Core Concept	Important Evidence	Interview Focus
Q1	Retail association mining	Repeated product combinations	Business application
Q2	Apriori vs FP-Growth	Candidate generation vs FP-tree	Method selection
Q3	Support and confidence	40% support for A,B; 50% confidence A→B	Correct calculation
Q4	Multilevel rules	Electronics 100%, Mobile 60%, Laptop 40%	Hierarchy comparison
Q5	Scalable mining	Partitioning, parallelism, pruning, compact representation	Large-data challenges

Final Interview Guidance

For the mock interview, the strongest response pattern is: answer the question directly, use the supplied dataset as evidence, explain the reasoning, and finish with a practical implication. This structure supports the assessment criteria without adding unrelated technical material.

Conclusion

The assessment responses address all five questions from the uploaded Assessment Tool 3 and are organized to demonstrate the six evaluation areas: technical knowledge, problem solving, analytical thinking, communication, confidence and response quality. The colour diagrams are included only where they improve conceptual understanding: retail rule discovery, Apriori versus FP-Growth, support/confidence, the multilevel hierarchy and scalable mining.